

Code:

```
#include <stdio.h>
int main() {
    int n;
    if (scanf("%d", &n) != 1) {
        return 0;
    }
    int array[n];
    for(int i=0;i<n; i++) {
        scanf("%d", &array[i]);
    }
    // Step 1: Sort the array in ascending order
    for (int i = 0; i < n - 1; i++) {
        for (int j = i + 1; j < n; j++) {
            if(array[i]>array[j]) {
                int temp = array[i];
                array[i] = array[j];
                array[j] = temp;
            }
        }
    }
    // Step 2: Swap adjacent elements
    for (int i = 0; i < n - 1; i += 2) {
        int temp = array[i];
        array[i] = array[i + 1];
        array[i + 1] = temp;
    }
    // Output the wave array
    for (int i = 0; i < n; i++) {
        printf("%d ", array[i]);
    }
    printf("\n");
    return 0;
}
```

Input:

```
6
3 6 5 10 7 20
```

Output:

```
5 3 7 6 20 10
```